

MUHAMMED MAHMUD

Robotics & Mechatronics Engineering Student

Autonomous Systems | Embodied AI | Swarm Intelligence | SLAM | UAV Systems

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PROFILE SUMMARY

Robotics and Mechatronics Engineering student focused on autonomous systems, embodied AI, and multi-agent robotics. Currently contributing to embodied AI and industrial humanoid development in applied R&D at Flolabs, and conducting UAV swarm research at KFUPM. Founder of CityWeave, applying a UAV swarm exploration framework (AVCE) to city-scale 3D mapping. Experienced in decentralized navigation, SLAM-based mapping, and simulation-first validation of autonomous systems, with long-term plans to pursue graduate studies in AI and Robotics.

PROFESSIONAL EXPERIENCE

CityWeave — Founder

2026 – Present

Startup applying the AVCE (Active Viewpoint-Coupled Exploration) UAV swarm framework to city-scale 3D mapping and urban planning.

- Leading development of the AVCE framework, reaching 88.2% simulated environment coverage; core IP with a patent filing in progress.
- Prepared and pitched investor materials, including a seed-round pitch targeting \$750K in funding.

Flolabs — AI & Robotics Intern

Jan 2026 – Present

IoT Team (CAIPO Project) and Robotics & Humanoid Team

- Contributing to the CAIPO embodied AI project within the IoT team, supporting development of intelligent connected systems that integrate perception and decision pipelines.
- Built the CAIPO wearable device LED state management library (17 states, priority system, base-state architecture) with a Python GUI companion for an ESP32-S3 platform.
- Member of the robotics team developing an industrial humanoid robot from scratch, contributing to system integration, autonomy components, and testing workflows.
- Supporting sensor integration, control validation, and iterative prototyping of humanoid subsystems, with structured experiment logs for reproducibility.

KFUPM — UAV Swarm Researcher

2026 – Present

Supervised by Professor Ayman Muhammad Abdallah

- Conducting research on decentralized UAV swarm coordination and exploration, forming the technical basis for the AVCE framework.
- Designing and validating multi-agent simulation environments for cooperative navigation and mapping.

Akiyam — AI Research Intern (Game Development)

Jan 2026 – Present

- Researched applications of machine learning and AI in game development, including behavioral modeling, agent intelligence, and adaptive systems.
- Investigated algorithmic approaches for intelligent NPC behavior and dynamic decision systems, analyzing performance and scalability of AI-driven gameplay mechanics.

Arc Technologies — Machine Learning Intern

Jan 2026 – Present

- Studied core machine learning algorithms and optimization techniques; implemented and evaluated supervised learning models across multiple datasets (spam classification, California Housing, Iris).
- Strengthened mathematical and computational foundations for AI systems development.

EDUCATION

B.Eng. Robotics & Mechatronics

Expected Graduation: July 2027

Belgorod State National Research University (НИУ «БелГУ»), Russia — Group 12002312

- Aurora Scholarship recipient.
- Relevant coursework: Robotics Systems, Control Systems, Mechatronics, Embedded Systems, Signals & Systems, Engineering Programming, Applied Mechanics, 3D Modelling.

SELECTED PROJECTS

AVCE UAV Swarm Simulation & 3D Mapping Pipeline

- Built a PyBullet + ROS 2 multi-UAV simulation using decentralized, bio-inspired swarm rules, reaching 88.2% environment coverage in 64 steps.
- Began Phase 3 reinforcement learning training using Stable-Baselines3 (PPO) with a flattened Dict observation wrapper.
- Aggregating distributed agent point-cloud data into a unified 3D map; analyzing localization drift and reconstruction accuracy.

ROS 2 Heterogeneous Multi-Robot Autonomy Architecture

- Designed and implemented a modular ROS 2 autonomy architecture for heterogeneous multi-robot coordination and collaborative 3D mapping.
- Developed distributed perception pipelines and inter-robot communication strategies for decentralized data fusion.
- Tech: ROS 2, Python/C++, Gazebo, point-cloud processing.

4-DOF Belt-Drive Robotic Arm (in progress)

- Designing a first hardware manipulator build with ~1–2 kg payload, NEMA 23/17 steppers, GT2 belt reductions, and an ESP32-S3 controller with TMC2209 drivers.
- FDM-printed structure with a wiring and safety manual already drafted.

Micro-Drone Swarm Platform

- Built an indoor swarm platform around ESP32-S3, DRV8833 dual H-bridges, PMW3901 optical flow, and VL53L1X ToF sensors using ESP-Drone firmware.
- Targeting multi-unit indoor swarm experiments controlled via cflib/Python.

Applied ML & Computer Vision Toolkits

- Email spam classifier (Python, Scikit-learn), computer vision toolkit for classical image processing (OpenCV), and a real-time face mask detection system using CNNs.

TECHNICAL SKILLS

Programming & Tools: Python, C++, ROS 2, Linux (Ubuntu), Git/GitHub

Robotics & Autonomy: Multi-agent systems, swarm algorithms (Boids), autonomous navigation, collision avoidance, SLAM fundamentals, path planning (A*, RRT)

Simulation: Gazebo, PyBullet, CoppeliaSim, Isaac Sim (learning), Unity/Unreal, OpenCV, Open3D

Machine Learning: Supervised learning, reinforcement learning (PPO/Stable-Baselines3), model evaluation, optimization fundamentals

CAD & Design: FreeCAD, 3D Modelling, URDF/Xacro

Embedded Systems: ESP32-S3, Arduino, MSP430, TMC2209/DRV8833 motor drivers, sensor integration (LiDAR, ToF, optical flow)

LEADERSHIP & ACTIVITIES

Association of Nigerian Scholars in Russia (ANSSIR), Belgorod

2024 – 2026

- Head of Media Team (2025–2026); previously Member of Media Team and Social Committee (2024–2025).
- Built and grew the association's social media presence and led public relations efforts.
- Created a structured student database for the association.

AWARDS

Aurora Scholarship — Merit-based scholarship recipient

BEA Scholarship Winner — Nigerian Federal Government, Oct 2022

Honor Student / Face of the Faculty — Kursk State University, May 2023

Winner, Russian Language Olympiad